

Complex Analysis

Selected Topics on Mathematical Physics | Autumn 2026

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I Prerequisites

Students are expected to be familiar with the following topics:

- Elementary set theory
- Elementary calculus

II Course Structure

The course meets twice per week for lectures. In addition, a weekly tutoring session is held to support work on the problem sheets; attendance is strongly encouraged, as the sheets form an integral part of the course.

	Format	Duration
Regular sessions	Theory and derivations	80 minutes
Tutoring session	Problem sheet workshop	90 minutes

Problem sheets should be attempted *before* the tutoring session. Hints and solutions are discussed in the tutoring session and collected shortly thereafter.

III Course Description

Complex analysis is one of the most beautiful and surprising areas of mathematics: a theory that begins with the simple act of allowing the square root of -1 and ends by revealing a hidden order beneath the surface of calculus. This course develops the theory of functions of a complex variable with full mathematical rigour. We will follow the ideas from their classical origins to their modern form: complex differentiation and the Cauchy–Riemann equations, contour integration and Cauchy’s integral theorem, the spectacular consequences of Cauchy’s formula, the classification of singularities, and the residue theorem as a powerful engine for computing integrals that resist every real-variable technique. Along the way, students will encounter results that are, without exaggeration, among the most elegant in all of mathematics.

Beyond its intrinsic beauty, complex analysis is indispensable: it underlies signal processing and the Fourier transform, quantum mechanics and scattering theory, analytic number theory and the Riemann zeta function, fluid dynamics, and control theory. Students who complete this course will have acquired both a powerful set of computational tools and a refined sense of mathematical elegance that will inform their work across virtually every advanced discipline.

This course was originally intended as a replacement for a traditional first course on introduction to real analysis, so many topics are part of such course.

IV Contents

1. **The Real Body** — Axiomatic foundations for the set $\mathbb{N}$ of natural numbers; construction of the sets $\mathbb{Z}$ and $\mathbb{Q}$ of naturals and integers and rationals via equivalence classes; construction of the set $\mathbb{R}$ of real numbers via Cauchy sequences; real arithmetic.
2. **A Glimpse into Abstract Algebra** — Algebraic structures; group definition, Abelian groups; ring definition, rings with unity, commutative rings, division rings, bodies.
3. **The Complex Body** — Axiomatic foundations for the field $\mathbb{C}$; complex arithmetic.
4. **Metric Topology of $\mathbb{C}$** — Characterisation of subsets of $\mathbb{C}$; interior, exterior, boundary, closure of a set; geometric properties of complex arithmetic; complex sequences; the Riemann's sphere and the point at infinity ∞ ; convergent sequences; Cauchy sequences.
5. **Limits and Continuity** — Limit of a complex function; limit theorems; extension of the limit concept; continuity for complex functions; characterisation of continuous maps via open subsets.
6. **Infinite Series** — Formal series; properties of series; tests for absolute and nonabsolute convergence; limit superior and limit inferior; Cauchy's strong n th root test.
7. **Sequences and Series of Functions** — Series of functions; power series; Weierstrass criterion for uniformly convergent sequences; radius of convergence.
8. **The Derivative** — Complex differentiability; holomorphic and analytic functions; differentiation rules; the chain rule; differentiation of series; the Cauchy-Riemann equations.
9. **The Exponential Function** — Analytic definition of the exponential function; Taylor's theorem for real variable; extension to the complex exponential function; the trigonometric functions; polar representation of a complex number; the n th roots of unity.
10. **The Complex Integral** — The Henstock-Kurzweil integral for real functions; improper and Lebesgue integration; the fundamental theorem of calculus; extension of the Henstock-Kurzweil integral to contour integration; the fundamental theorem of contour integration; Cauchy-Goursat's theorem; Morera's theorem; Cauchy's integral formula; Leibniz rule; the fundamental theorem of Algebra.
11. **Residue Theory** — Singularities; Laurent series; Cauchy's residue theorem; poles of a function; Jordan's lemma for contour integration; Cauchy's principal value; applications of residue theory to real integrals.
12. **Applications of Residues** — Real integrals via complex methods; Dirichlet integrals; integrands of the form $F(\cos \theta, \sin \theta)$; integrals of quotients of polynomials; the generalised Gaussian integral.

This syllabus is subject to change. Any modifications will be announced in class.

V References

The star rating reflects the mathematical sophistication of each text: (★) accessible with the course prerequisites; (★★) requires additional mathematical maturity; (★★★) intended for readers revisiting the subject — not recommended if this is your first encounter with the material.

- [1] Bartle, R. G. and Sherbert, D. R. *Introduction to Real Analysis*. 4th ed. Wiley, 2011. [Primary (★)]
- [2] Asmar, N. H. and Grafakos, L. *Complex Analysis with Applications*. Springer, 2018. [Primary (★★)]
- [3] Tao, T. *Analysis I*. 4th ed. Texts and Readings in Mathematics, vol. 37. Hindustan Book Agency / Springer, 2022. [Supplementary (★)]
- [4] Hassani, S. *Mathematical Physics: A Modern Introduction to Its Foundations*. 2nd ed. Springer, 2013. [Background reference (★★★)]
- [5] Johar, S. *The Big Book of Real Analysis*. Springer, 2023. [Background reference (★★)]
- [6] Presilla, C. *Elementi di Analisi Complessa: Funzioni di una Variabile*. Springer, 2014. [Background reference (★★)]